MIL-S-1222H
INT. AMENDMENT 2(SH)
12 October 1990
USED IN LIEU OF
AMENDMENT 1
31 March 1989

MILITARY SPECIFICATION

STUDS, BOLTS, HEX CAP SCREWS, SOCKET HEAD CAP SCREWS, AND NUTS

This interim amendment is approved for use within the Naval Sea Systems Command, with MIL-S-1222H, dated 21 October 1986.

PAGE 1

* 1.1, second line: Delete, "in diameters of 1/4 inch and larger."

1.2, under "Studs, type IV": After "Continuous thread", add "(Bolt-stud)".

PAGE 2

2.1.1, under "SPECIFICATIONS, FEDERAL, QQ-C-465": Delete "Numbers 606, 614, 630, 632M, and 642" and substitute "Numbers C60600, C61400, C63000, C63200, and C64200".

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* 2.1.1, under "SPECIFICATIONS, MILITARY": Delete "MIL-N-24106".

* 2.1.1, under "STANDARDS, FEDERAL": Delete "FED-STD-H28/20"; Add, "FED-STD-H28/2 - Screw Thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes, and General Applications."

2.1.1, under "STANDARDS, MILITARY": Add "MIL-STD-792 - Identification ~~Marking Requirements for Special Purpose Components.~~"

* 2.2, under "ANSI": Add: "B1.3 - Screw Thread Gaging Systems for Dimensional Acceptability."

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2.2, under "ASTM": Delete reference to "ASTM A 751".

2.2, under "SAE" after J 995, add: "AMS 2750 - Pyrometry."

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* 3.1, line 3, above TABLE I: Add:

"WARNING NOTE: STRENGTH LEVELS FOR MARTENSITIC STAINLESS STEELS (GRADES 410, 416, AND 416Se) CONDITION H, OF MIL-S-1222H DO NOT HAVE STRENGTHS EQUIVALENT TO CONDITION H OF MIL-S-1222G. HOWEVER, CONDITION H OF MIL-S-1222H WITH INTERIM AMENDMENT 2 IS EQUIVALENT TO CONDITION H OF MIL-S-1222G."

Table I, under "Material Grade" for alloy steels: Delete "574" and substitute "A574".

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* Table I, under "Identification Marking" for martensitic corrosion resistant steels: Delete footnote "5/"; replace 410H, 416H, and 416SeH, with 410, 416, and 416Se; replace 410HT, 416HT, and 416SeHT with 410, 416, and 416SE.

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* Table I, under "Identification Marking" for martensitic corrosion resistant steels: Delete footnote "5/"; replace 410H, 416H, and 416SeH with 410, 416, and 416Se; replace 410HT, 416HT, and 416SeHT with 410, 416, and 416SE.

Table I, under chemical requirement for Ni-Cu-Al: Delete "class A".

Table I, under "Identification Marking" for Ni-Cu-Al nut: Add "8/".

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Table I, under "Material grade" for copper alloys, Nickel-aluminum bronze: Delete "632" and substitute "C63200".

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Table I, footnote 4: Delete and substitute:

"4/ In addition to the material grade, the marking shall include the heat treatment symbol "An" when the fastener is machined from annealed barstock or when the fastener is annealed after forming or the identification symbol "SH" when the fastener is cold headed and rolled from strain hardened stock thus acquiring a degree of cold work (see tables II, III, and IV)."

Table I, footnote 6, (f): Delete "651".

Table I, footnote 8: Delete and substitute: "See 3.9."

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3.1.2.1(a): Delete first sentence and substitute: "When fasteners are produced by cold heading and reduction of the shank, the starting material shall be in the cold drawn and annealed, hot finished and annealed, or hot forged and annealed condition."

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3.1.2.1, Add new (d) as follows:

"(d) When fasteners are machined from annealed barstock, threading may be by cutting or rolling. The finished fasteners shall be reannealed and age hardened."

3.1.2.2, Delete and substitute:

"3.1.2.2 Grade 400 externally threaded fasteners. Grade 400 externally threaded fasteners shall be fabricated from either hot finished barstock or cold drawn and stress relieved material in accordance with QQ-N-281. When specified (see 6.2.1), fasteners shall be fabricated from cold drawn material stress relieved at 900 to 1200°F. Heat treatment shall be controlled in accordance with AMS 2750. Tensile properties shall be determined after heat treatment."

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Table II, under "Nominal diameter" for carbon and alloy steels and 300 series corrosion resistant steels: Delete "1/4 inch to *" wherever it appears and substitute "* or less". (Where * is a variable number.)

Table II, under "Nominal diameter" for 400 series corrosion resisting steels: Delete "1/4 inch and over" and substitute "All sizes."

Table II, Delete lines covering 300 series corrosion resisting steel and insert new lines as shown.

			Machined specimens from full size fasteners			
Grade	Condition	Diameter	Tensile strength ksi	Yield strength ksi min	Proof stress ksi min	Rockwell hardness
303, 303Se 304, 305	Annealed "An"	All sizes	75-100	30	---	B95 max
316, 321 347, 384	Strain hardened "SH"	Less than 3/4	120-160	95	---	---
		3/4 to 1	110-150	75	---	---
		1-1/8 to 1-1/4	100-140	60	---	---
		1-3/8 to 1-1/2	95-130	45	---	---

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Machined specimens from fasteners or on parent barstock			
Tensile strength ksi min	Yield strength ksi min	Elongation in 4D percent min	Reduction of area percent min
75	30	20	---
115	90	12	---
105	70	15	---
95	55	20	---
90	40	28	---

* Table II, For corrosion resistant steels; under grade: Delete "431"; under "Heat treatment or condition" for 410, 416, 416Se: Delete "H" and substitute "T"; Delete "HT" and substitute "H". Add new line as shown:

Grade	Condition	Diameter	Machined specimens from full sized fasteners: Tensile strength ksi	Yield strength ksi, min	Proof stress ksi, min	Rockwell hardness
410, 416, 416Se	T	All sizes	125-150	95	---	C25-34
	H	All sizes	180-220	135	---	C38-47
431	H	All sizes	125-150	95	---	C25-34
	HT	All sizes	180-220	135	---	C38-47

Machined specimens from fasteners or parent barstock			
Tensile strength ksi, min	Yield strength ksi, min	Elongation in 4D percent min	Reduction in area percent min
125	95	20	---
180	135	12	---
125	95	20	---
180	135	12	---

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Table II, under "Nominal diameter" for copper and aluminum alloys and grade 630 stainless steel: Delete "1/4 and over" and substitute "All sizes".

* Table II, under "Nominal diameter for nickel alloys 400 and 405: Delete "1/4 and over" and substitute "All sizes".

Table II, under "Rockwell hardness" for grade 500 nickel alloy: Delete "C24 min" and substitute "C24-35" for all sizes.

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* Table II, under "Nominal diameter" for titanium alloys: Delete "1/4 to *" wherever it appears and substitute "* or less".

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* Table III, under "Nominal diameter" for all grades: Delete "1/4 to *" wherever it appears and substitute "All sizes".

Table III, under "Rockwell hardness" for grade 500: Delete "C24 min" and substitute "C24-35" for all sizes.

* Table III, under "Grade" for grades "574, 4340": Delete: "574 and substitute A574.

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* Table IV, under "Nominal size" for all grades: Delete "1/4 and over" wherever it appears and substitute "All sizes".

* Table IV, under "Grade": Delete "431", under "Heat treatment or condition" for corrosion resistant steels 410, 416, 416Se: Delete "H" and substitute "T"; Delete: "HT" and substitute "H". Add new line as follows:

"431	H	all sizes	125	C25-34
431	HT	all sizes	180	C38-47"

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3.8.1, Delete first sentence and substitute:

"Except for stud end threads of style a or b studs, screw threads shall be in accordance with ANSI B1.1 and unless otherwise specified (see 6.2.1), shall be unified class 2A/2B."

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Table VI, under, "Full body, Type I," "Styles a and c" and Styles b and d, maximums for both 1-3/8 sizes: Delete "1.295" and substitute "1.395".

Table VI, footnote 1/, line 2: Delete "FED-STD-H28/20" and substitute "FED-STD-H28/2".

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3.8.2.4, under "Tolerance for nominal stud diameter": Delete "5/8 to 1-1/4" and substitute "9/16 to 1-1/4".

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* 3.9, Delete and substitute:

"3.9 Identification marking. Unless otherwise specified (see 6.2.1), identification symbols/marking shall be as shown in table I. When heat to heat traceability (lot definition "a") is desired, and for grade 500 nickel alloy fasteners, the material symbol, manufacturer's symbol and the lot number shall be marked on the fasteners greater than 1/4 inch nominal size. For fasteners 1/4 inch nominal size or less, if all of the marking on the item cannot be applied due to space limitations, the marking shall be applied using the following order of precedence: material symbol, lot number, manufacturer's symbol. The material symbol is mandatory for all fasteners. For purposes of this document, hardened material shall be defined as that with a maximum hardness in excess of HRC 35. Marking shall be applied to the head of bolts, hex cap screws and socket head cap screws, the nut end of type I, II, III, studs, either end of type IV studs and the top face of nuts. Marking shall be permanent and in accordance with MIL-STD-792.

3.10 and 3.10.1, Delete and substitute:

"3.10 Workmanship. Fasteners shall be uniform in quality and condition, free from rust, scale, cracks, seams, bursts, voids, laps, nicks, gouges, and burrs as determined by the applicable nondestructive tests."

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Table XI, under "Type of test or inspection": Delete "Nondestructive testing" from the box with "Permeability measurement."

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* 4.4.3, Delete third sentence and substitute: "Screw thread gaging shall be variables/indicating in accordance with ANSI B1.3, System 22. Screw thread gauges shall be in accordance with ANSI B1.2."

* 4.4.1, line 1: Delete "with threads formed by upsetting".

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Table XIII, footnote 4: Delete.

Table XIII, footnote 5: Delete and substitute: "4/ See 4.4.4.4."

4.4.4.4 Add new second and third sentences:

"Nuts with a proof load greater than 150,000 pounds shall not be subject to the proof load test but shall meet the hardness test of 4.4.4.5. For those facilities with the capability, proof load test results shall be submitted along with hardness test results."

At end of new fifth sentence: Delete "Table IV" and substitute "Table V".

4.4.4.5, line 2: Delete "ASTM A606" and substitute "ASTM F606".

4.4.4.5, line 2: Delete "table II and III" and substitute "tables II, III, and IV".

4.4.4.6, Delete and substitute:

"4.4.4.6 Alternate mechanical properties. Certified mechanical test reports (see 4.5) furnished by the barstock supplier which demonstrates compliance with the mechanical properties of tables II, III, and IV may be substituted for mechanical testing of fasteners provided that: (a) lot definition "a" is employed, (b) marking or identification is maintained to ensure traceability to the barstock certification, (c) no further heat treatment is performed, and (d) fabrication is solely by machining or with thread forming by machining or rolling. The following exceptions apply:

(a) For bolts and screws fabricated by thread rolling and cold heading, the wedge tensile test of 4.4.4.1 shall be required in addition to the barstock suppliers certification.
(b) For grade 500 nickel-copper-aluminum alloy studs manufactured from age hardened barstock, only thread forming by cutting shall be permitted.

4.4.5, Delete and substitute:

"4.4.5 Nondestructive testing. Fasteners shall be nondestructively inspected by wet magnetic particle or liquid penetrant test in accordance with MIL-STD-271. Fasteners made from magnetic materials shall be wet magnetic particle inspected. All other fasteners shall be inspected by the liquid penetrant test. When protective coating is specified, the nondestructive tests shall be performed before the coating process.

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* 4.4.8: Delete "574" and substitute "A574".

4.4.10, Insert "be" between "shall" and "impact".

4.5, Delete "When specified in the contract or order," and substitute "when specified (see 6.2.1),".

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Table XIV, under "Steel grades": Delete "Carbon steel grades SAE 2 and 5" and substitute "Carbon steel grades SAE 2, 5, and 8".

Table XIV, under "Steel grades": Delete "Grade B7 externally threaded used with grade 2H, 4, or 7 nuts" and substitute "Grade B7 externally threaded fasteners used with grade 2H, 4, or 7 nuts."

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6.1.4, line 1: Delete "steal" and substitute "steel".

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6.2.1, (b): Delete and substitute:

"(b) Type, style, material grade, and heat treatment or condition of fasteners (see 1.2, 3.1, 3.2, and tables I, II, III, and IV)."

* 6.2.1, (c): Delete and substitute:

"(c) If grade 400 should be cold drawn and stress relieved at 900 to 1200 F (see 3.1.2.2).

6.2.1: Add new (l): "(l) Test report, if required (see 4.5)."

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* 6.2.2: Add:

"Paragraph no.	Data requirements title	Applicable DID no.	Option
4.5	Certification data for non-nuclear level I material"	DI-MISC-80705	----

6.3, line 3: Delete "table XVI" and substitute "Table XVII".

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Table XVI, title: Delete "Table XVI" and substitute "Table XVII".

Table XVI, under "Description" for studs: After "Continuous thread", add "(bolt-stud)".

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6.4, line 3: Delete "6.4.1 and 6.4.2" and substitute "6.4.1, 6.4.2, and 6.4.3".

6.4, lines 3 and 5: Delete "table XVI" and substitute "table XVII".

6.4.1, line 12: Delete "table XVII" and substitute "table XVIII".

Table XVII, title: Delete "Table XVII" and substitute "Table XVIII".

Table XVII, under "MIL-S-1222H" for carbon and alloy steels: Add "A574" and "4340".

Table XVII, under "MIL-B-857" for corrosion resistant steel: Add "304".

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Table XVII, title: Delete "Table XVII" and substitute "Table XVIII".

Table XVIII, under "MIL-S-1222H" for copper alloys: Delete "632" and substitute "C63200".

Table XVIII, under "MIL-B-857A" for copper alloy 544: Delete "Aluminum bronze" and substitute "Phosphor bronze".

Table XVIII, under "MIL-S-1222H, MIL-S-001222G and MIL-B-857A" for copper alloys: Add "670", "670", and "Manganese bronze", respectively.

Table XVIII, under "MIL-B-857A" for nickel alloy 500: Delete "Ni-Cu-A2" and substitute "Ni-Cu-A1".

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6.6: Add: "Studs".

6.7: Delete and substitute:

"6.7 Changes from previous issue. The margins of this interim amendment are marked with an asterisk (*) to indicate where changes from amendment 1 were made. This was done as a convenience only and the Government assumes no liability for

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any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the amendment 1.

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